Wireless Communication

Indu Bhardwaj
Bharati Vidyapeeth College of Engineering.
New Delhi, India
indubhardwaj2011@gmail.com

UNIT – I

Introduction To Wireless Communication Systems:

Evolution of mobile radio communications; examples of wireless comm. systems; paging systems; Cordless telephone systems; overview of generations of cellular systems, comparison of various wireless systems.

Introduction to Personal Communication Services (PCS):

PCS architecture, Mobility management, Networks signaling. basic cellular system, multiple access techniques: FDMA, TDMA, CDMA.

Introduction to Wireless Channels and Diversity:

Fast Fading Wireless Channel Modeling, Rayleigh/Ricean Fading Channels, BER Performance in Fading Channels, Introduction to Diversity modeling for Wireless Communications

Suggested Readings

Text Books:

- Raj Pandya, “Mobile & Personnel communication Systems and Services”, Prentice Hall India, 2001.
- Theodore S. Rappaport, “Wireless Communication- Principles and practices,” 2nd Ed., Pearson Education Pvt. Ltd, 5th Edition, 2008.

Reference Books:

- T.L.Singhal “Wireless Communication”, Tata McGraw Hill Publication.
- Jochen Schiller, “Mobile communications,” Pearson Education Pvt. Ltd., 2002.
- Yi –Bing Lin & Imrich Chlamatac, “Wireless and Mobile Networks Architecture,” John Wiley & Sons, 2001.
- Lee, W.C.Y., “Mobile Cellular Telecommunication”, 2nd Edition, McGraw Hill,1998.
- Smith & Collins, “3G Wireless Networks,” TMH, 2007
- Schiller, Jochen, “Mobile Communications”, 2nd Edition, Addison Wesley

PCS (Personal Communication Services)

- PCS (Personal Communication Services) includes wide variety of network services that includes wireless access and personal mobility services
- PCS is a general term for a variety of voice, data & multimedia services. (GSM is a kind of PCS system)
- Such a Personal Communication Service can be accessed at any time, place or form (They are mobile/portable/ubiquitous).
- PCS lays the foundation architecture. GSM & GPRS build upon & extend the PCS architecture to achieve new functionality.
- Two of the most popular PCS are Cellular telephony Cordless and low-tier telephony

Several PCS systems

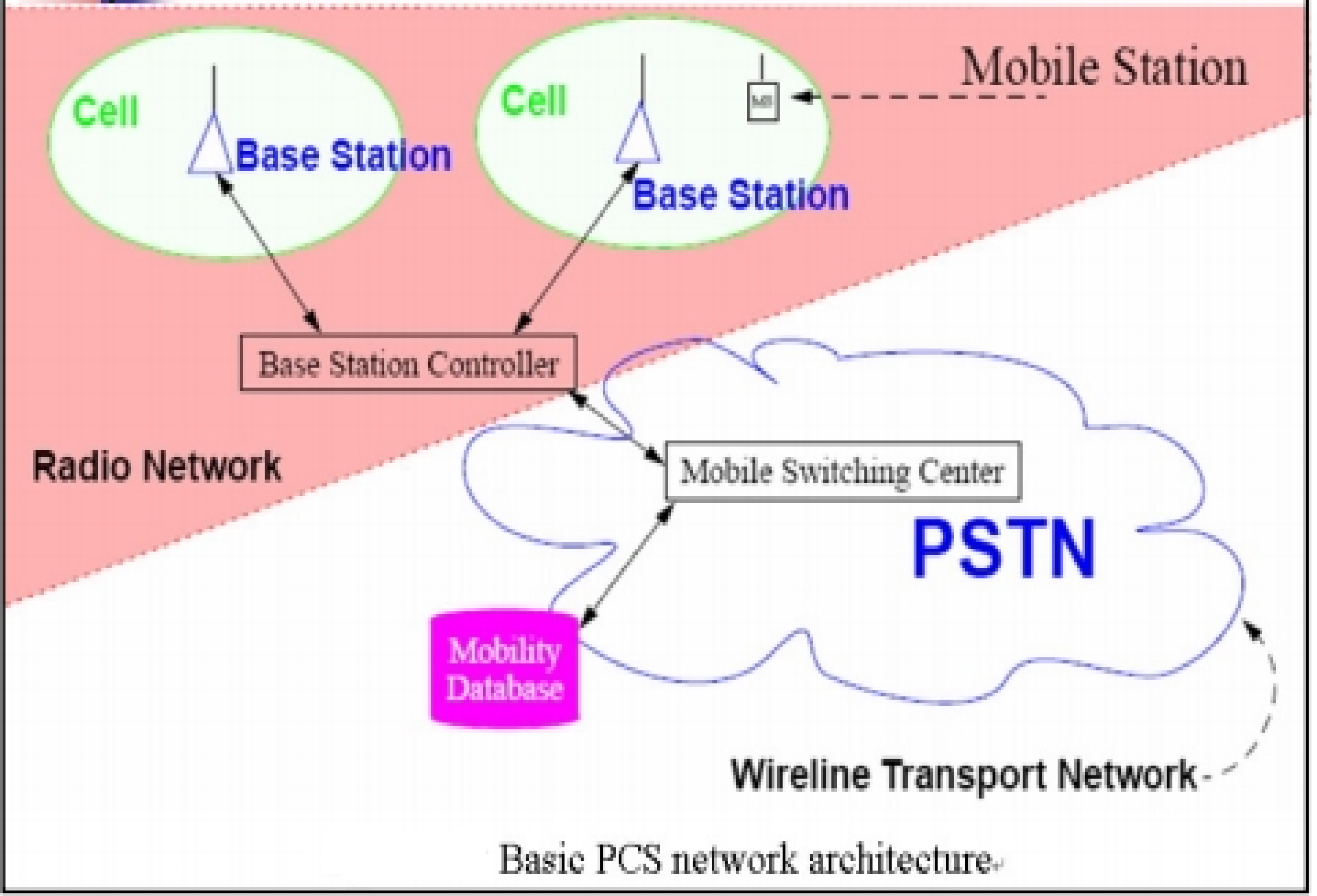
- **High-Tier Digital Cellular Systems** (mobile phone systems) □
  - Global System for Mobile Communication (GSM) □
  - IS-136 TDMA based Digital Advanced Mobile Phone Service (DAMPS) □
  - Personal Digital Cellular (PDC)
  - □IS-95 CDMA-based cdma One system
- **Low-Tier Telecommunication Systems** for residential, business, and public cordless access applications
  - □Cordless Telephone (CT2) □
  - Digital Enhanced Cordless Telephone (DECT) □Personal Access Communications Systems (PACS) □
  - Personal Handy System (PHS)

Several PCS systems

- **Wideband Wireless Systems** have been developed to accommodate Internet and multimedia
  - cdma2000 evolved from cdma One
  - W-CDMA proposed by Europe
  - TSCDMA proposed by China/Europe
- **Other PCS Systems**
  - Special data systems
    - CDPD: Cellular Digital Packet Data
    - RAM Mobile Data
    - Advanced Radio Data Information System (ARDIS)
  - Paging Systems
  - Mobile Satellite Systems
    - LEO, MEO, HEO satellites for data/voice
  - ISM band systems: Bluetooth, 802.11, etc.

PCS Architecture

- ❑The basic architecture consists of two parts.
  - Radio Network ❑
  - Wireline Transport Network



Basic PCS network architecture

Architecture- terminology

- Mobile Station
  - A station in the cellular radio service intended for use while in motion at unspecified locations. They can be either hand-held personal units (portables) or installed on vehicles (mobiles)
- Base station
  - A fixed station in a mobile radio system used for radio communication with the mobile stations. Base stations are located at the center or edge of a coverage region. They consists of radio channels and transmitter and receiver antennas mounted on top of a tower.
- Mobile Switching Center
  - Switching center which coordinates the routing of calls in a large service area. ☐The MSC is connected to the PSTN to provide services between the PCS users and the wireline users.

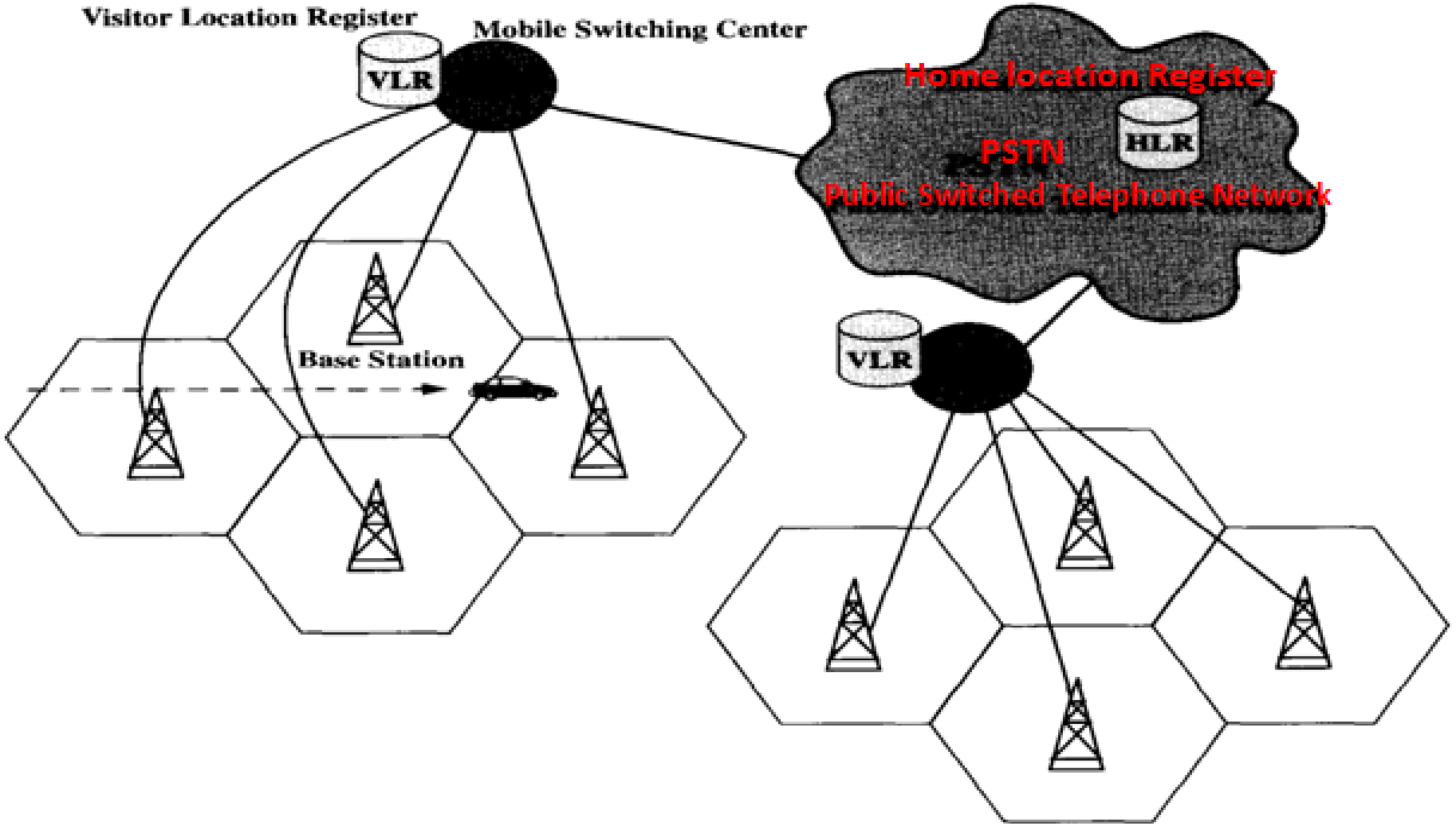
Architecture- terminology

- Control Channel
  - Radio channel used for transmission of call setup, call request, call initiation and other beacon and control purposes.
- Forward Channel
  - Radio channel used for transmission of information from the base station to the mobile
- Reverse Channel
  - Radio channel used for transmission of information from mobile to base station
- The Mobility Database connected to the MSC is used to track the locations of mobile stations.

PCS system architecture

- The mobile **service area** is covered by a set of **base stations (BSs)**, which are responsible for relaying the calls to and from the **mobile stations (MSs)** located in their coverage areas (or cells).
- The BSs are connected to **mobile switching centers (MSCs)** by land links.
- MSC
  - a telephone exchange configured specifically for mobile applications.
  - interfaces the MSs (via BSs) with the PSTN.
- Databases are used for roaming management:
  - **Home location register (HLR)**
  - **Visitor location register (VLR)**

Common PCS Network Architecture



Mobility Management in PCS

- Mobility is very important in mobile communication
- Mobility management is a defined as a way the network manages the movements of mobile users
- There are two aspects of mobility in a PCS network:
  - **Handoff**
  - **Roaming**

Handoff

- When a mobile user is engaged in conversation, the MS is connected to a BS via a radio link.

- If the mobile user moves from one BS to the coverage area of another BS, the radio link to the old BS is eventually disconnected, and a radio link to the new BS should be established to continue the conversation. This process is variously referred to as **automatic link transfer**, **handover**, **or handoff.**

Roaming

- When a mobile user moves from one PCS system (e.g., the system in New York City) to another (e.g., the system in Los Angeles), the system should be informed of the current location of the user. Otherwise, it would be impossible to deliver the services to the mobile user.
- To support mobility management, protocols such as EIA/TIA **Interim Standard 41** (**IS-41** or ANSI-41) or Global System for Mobile Communications (GSM) **Mobile Application Part (MAP)** have been defined for PCS networks.

Please NOTE

- **Handoff** - is carried out when mobile user moves from one BS to Other BS where as **Roaming** - is carried out when mobile user moves From one PCS system to other PCS system

- Now lets discuss handoff and roaming one by one for better understanding of mobility management

Handoff

- Three strategies have been proposed to detect the need for handoff:
  1. **mobile-controlled handoff (MCHO)**
  2. **network-controlled handoff (NCHO)**
  3. **mobile-assisted handoff (MAHO)**

Three strategies for handoff

1. **Mobile-Controlled Handoff (MCHO)-** The **MS** continuously monitors the signals of the surrounding BSs and initiates the handoff process when some handoff criteria are met. MCHO is used in DECT and PACS.
2. **Network-Controlled Handoff (NCHO)-** The surrounding **BSs** measure the signal from the MS, and the network initiates the handoff process when some handoff criteria are met. NCHO is used in CT-2 Plus and AMPS.
3. **Mobile-Assisted Handoff (MAHO)-** The network asks the **MS** to measure the signal from the surrounding BSs. The **network** makes the handoff decision based on reports from the MS. MAHO is used in GSM and IS-95 CDMA.

Two types of handoff

- The BSs involved in the handoff may be connected to the **same MSC** (**inter‑cell handoff** or **inter‑BS handoff**)

- The BSs involved in the handoff may be connected to two **different MSCs** (**inter-systemhandoff** or **inter‑MSC handoff**).

Inter-BS Handoff

- **The new and the old BSs are connected to the same MSC.**
- Assume that the need for handoff is detected by the MS; the following actions are taken:
  - 1. The MS momentarily suspends conversation and initiates the handoff procedure by signaling on an idle (currently free) channel in the new BS. Then it resumes the conversation on the old BS.
  - 2. Upon receipt of the signal, the MSC transfers the encryption information to the selected idle channel of the new BS and sets up the new conversation path to the MS through that channel. The switch bridges the new path with the old path and informs the MS to transfer from the old channel to the new channel.

Inter-BS Handoff

- 3. After the MS has been transferred to the new BS, it signals the network, and resumes conversation using the new channel.
- 4. Upon receipt of the handoff completion signal, the network removes the bridge from the path and releases resources associated with the old channel.

- This handoff procedure is used with the **mobile-controlled handoff strategy**

Inter-BS Handoff

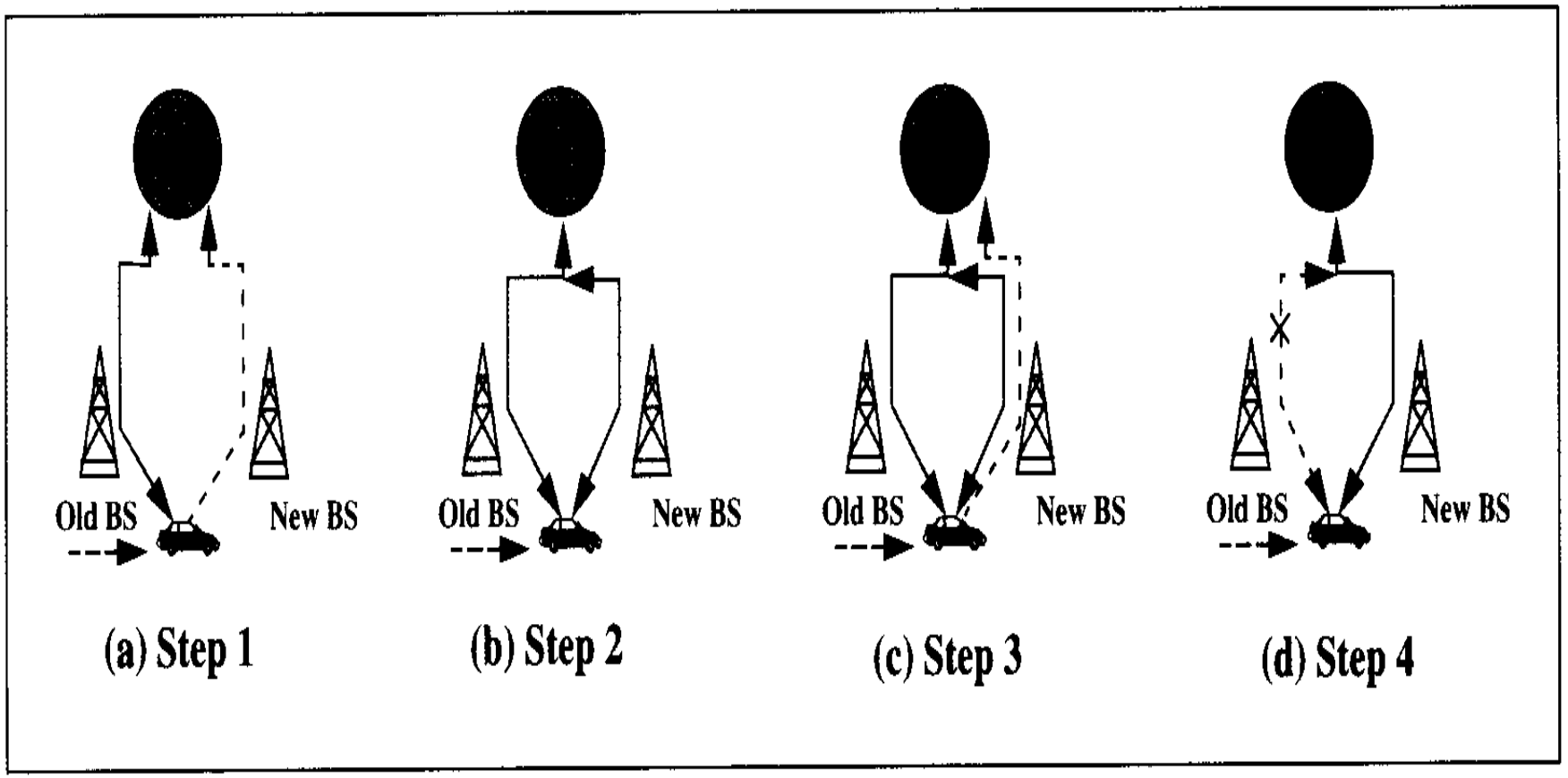


Figure 2.2 Inter-BS link transfer.

Inter-BS Handoff

- For the **network-controlled handoff strategy**, all handoff signaling messages are exchanged between the MS and the old BS though the failing link.

- The whole process must be completed as quickly as possible, to ensure that the new link is established before the old link fails.

Inter-BS Handoff

- If the new BS does not have an idle channel, the handoff call may be dropped (or forced to terminate).
- The **forced termination probability** is an important criterion in the performance evaluation of a PCS network.
- Forced termination of an ongoing call is considered less desirable than blocking a new call attempt.
- Most PCS networks handle a handoff in the same manner as a new call attempt. That is, if no channel is available, the handoff is **blocked** and the call is held on the current channel in the old cell until the call is completed or when the failing link is no longer available.
- This is referred to as the **non-prioritized scheme**.

Channel Assignment Schemes

- To reduce forced termination and to promote call completion, three channel assignment schemes have been proposed:
  1. **Reserved channel scheme**
  2. **Queuing priority scheme**
  3. **Subrating scheme**

1. Reserved channel scheme

- It is similar to the non-prioritized scheme, except that some channels in each BS are reserved for handoff calls.

2. Queuing priority scheme

- Adjacent coverage areas of BSs may overlapped.
- Thus, there is a considerable area where a call can be handled by either BS. This area is called the **handoff area**.
- If no channel is available in the new BS during handoff, the new BS buffers the handoff request in a waiting queue.
- The MS continues to use the channel with the old BS until either a channel in the new BS becomes available (and the handoff call is connected) or the MS moves out of the handoff area (and the call is forced to terminate).

Subrating scheme

- Creates a new channel for a handoff call **by sharing resources with an existing call if no channel is available in the new BS**.
- **Subrating** means an occupied full-rate channel is temporarily divided into two channels at half the original rate:
  - one to serve the existing call
  - and the other to serve the handoff request.
- When occupied channels are released, the subrated channels are immediately switched back to full-rate channels.

Inter-BS Handoff

- These handoff schemes can significantly reduce the probability of forced termination as well as the probability of call incompletion (new call blocking plus handoff call forced termination).

Intersystem Handoff

- In intersystem handoff, the new and old BSs are connected to two **different MSCs**.
- We trace the intersystem handoff procedure of IS-41 , where network-controlled handoff (**NCHO**) is assumed.
- In this figure, a communicating mobile user moves out of the BS served by MSC A and enters the area covered by MSC B.

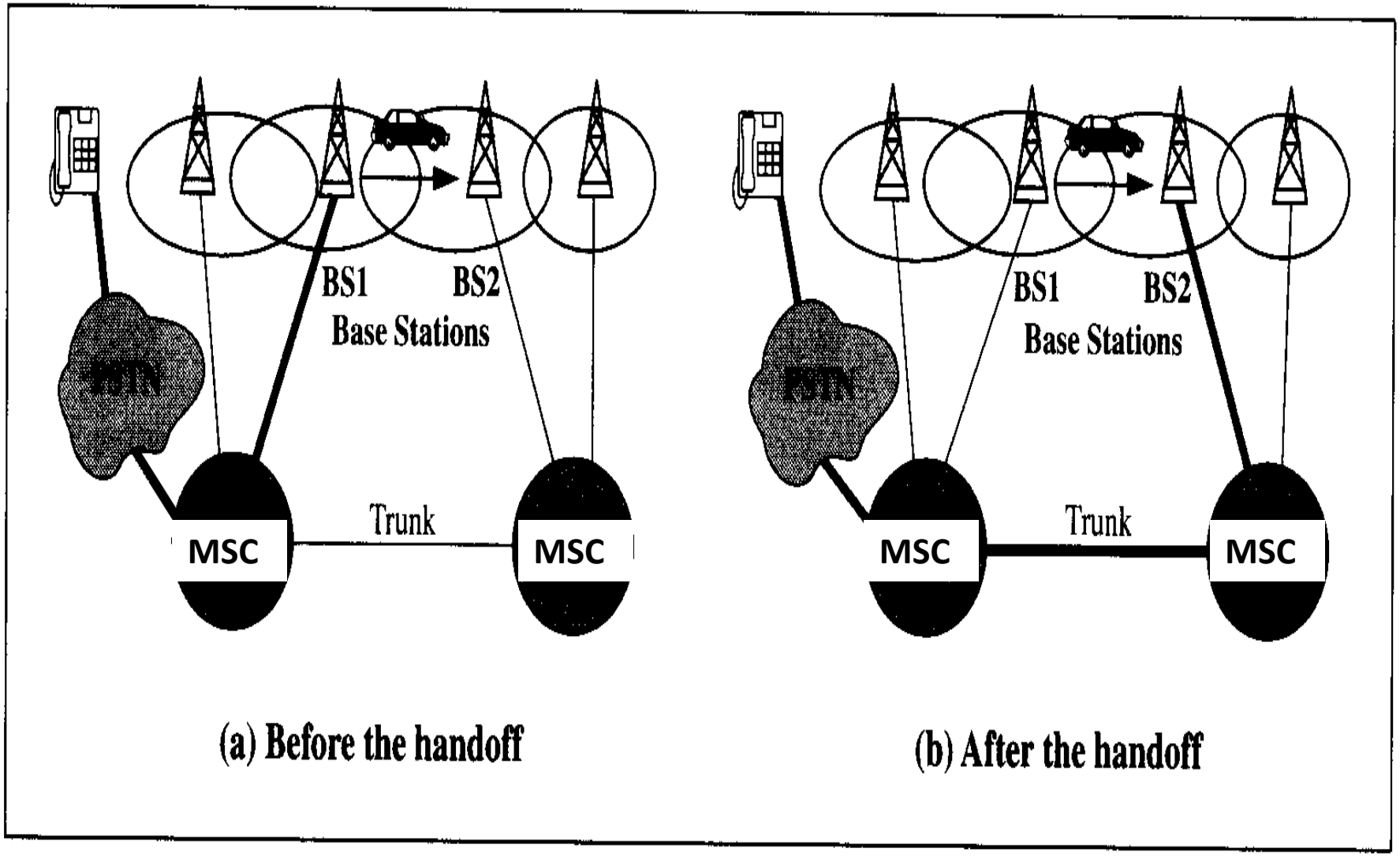


Figure 2.3 Intersystem handoff.

Intersystem Handoff

- Intersystem handoff requires the following steps:
- **Step 1**. MSC A requests MSC B to perform handoff measurements on the call in progress. MSC B then selects a candidate BS2 , and interrogates it for signal quality parameters on the call in progress. MSC B returns the signal quality parameter values, along with other relevant information, to MSC A.

Intersystem Handoff

- **Step 2.** MSC A checks if the MS has made too many handoffs recently (this is to avoid, for example, numerous handoffs between BS1 and BS2 a where the MS is moving within the overlapped area) or if intersystem trunks are not available. If so, MSC A exits the procedure. Otherwise, MSC A asks MSC B to set up a voice channel. Assuming that a voice channel is available in BS2, MSC B instructs MSC A to start the radio link transfer.

Intersystem Handoff

- **Step 3.** MSC A sends the MS a handoff order. The MS synchronizes to BS2. After the MS is connected to BS2, MSC B informs MSC A that the handoff is successful. MSC A then connects the call path (trunk) to MSC B and completes the handoff procedure.

Intersystem Handoff

- In this intersystem handoff process, MSC A is referred to as the **anchor MSC**, and is always in the call path before and after the handoff, as illustrated in the four cases in Figure 2.4.
- This anchor approach is used in all existing mobile phone networks because the re-establishment of a new call path (without involving MSC A) between MS and the new MSC would require extra trunk release/setup operations in PSTN, which is not available or is not cost-effective.

Intersystem Handoff

- If the MS moves back to MSC A again, the connection between MSC A and MSC B is removed (**handoff backward**).
- If the MS moves to the third MSC C, then MSC B will be in the call path (**handoff to third**).
- That is, the link between MSC B and MSC A is disconnected, and MSC C connects to MSC A directly.
- This process is called **path minimization**.

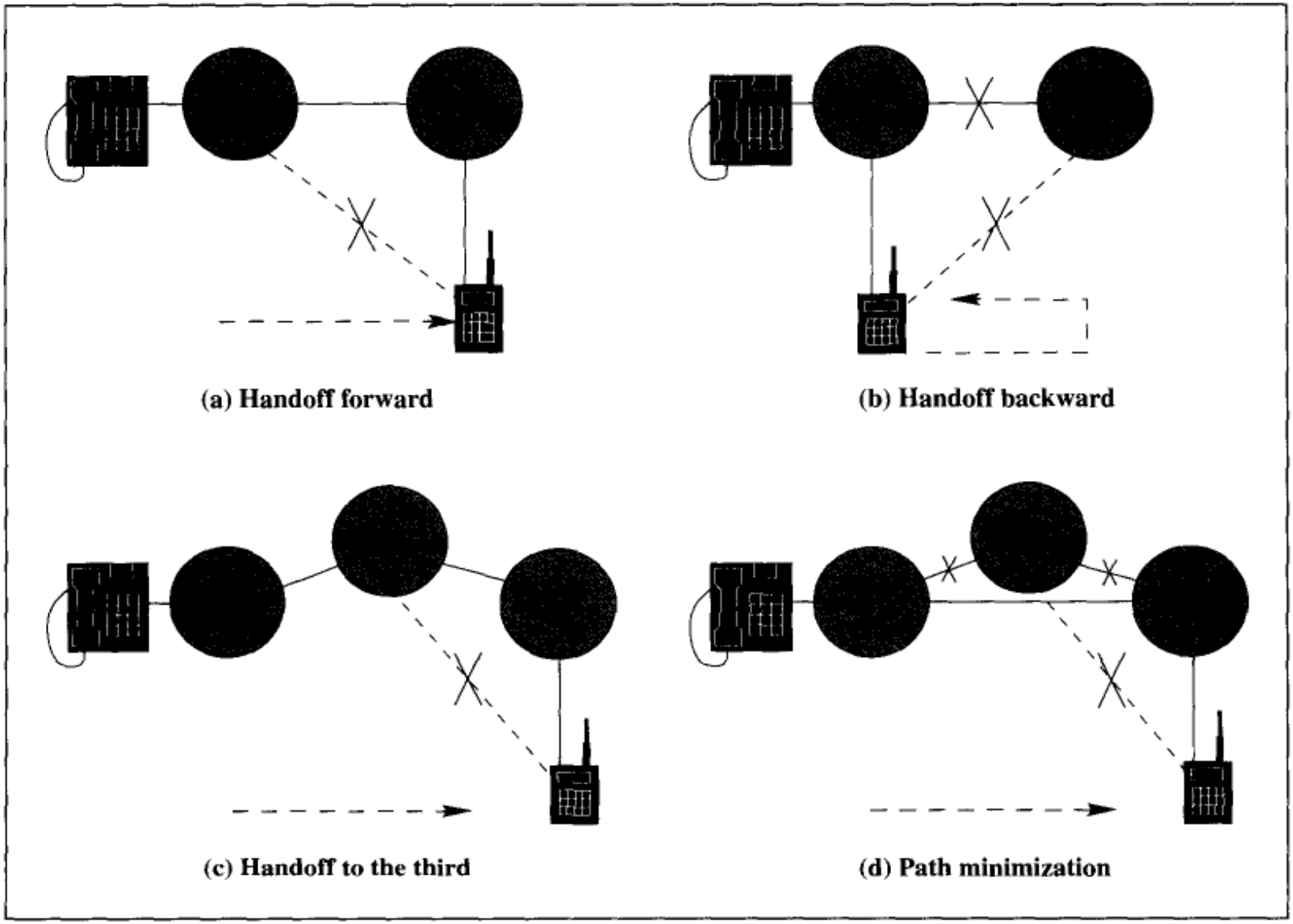


Figure 2.4 Handoff forward, handoff backward, and handoff to the third.

Roaming Management

- Two basic operations in roaming management are
  - **registration (or location update),** the process whereby an MS informs the system of its current location, and
  - **location tracking,** the process during which the system locates the MS. Location tracking is required when the network attempts to deliver a call to the mobile user.

Roaming Management

– The roaming management strategies proposed in the IS-41 and GSM MAP standards are two-level strategies in that they use a two-tier system of **home** and **visited databases**.

Home Location Register (HLR)

- When a user subscribes to the services of a PCS network, a record is created in the system's database, called the **home location register (HLR)**.
- The HLR is a network database that stores and manages all mobile subscriptions of a specific operator.
- Specifically, the HLR is the location register to which an MS identity is assigned for record purposes, such as directory number, profile information, current location, and validation period.

Visitor Location Register (VLR)

- When the mobile user visits a PCS network other than the home system, a temporary record for the mobile user is created in the visitor location register (VLR) of the visited system.

- The VLR temporarily stores subscription information for the visiting subscribers so that the corresponding MSC can provide service.

- In other words, the VLR is the "other" location register used to retrieve information for handling calls to or from a visiting mobile user.

Registration Procedure

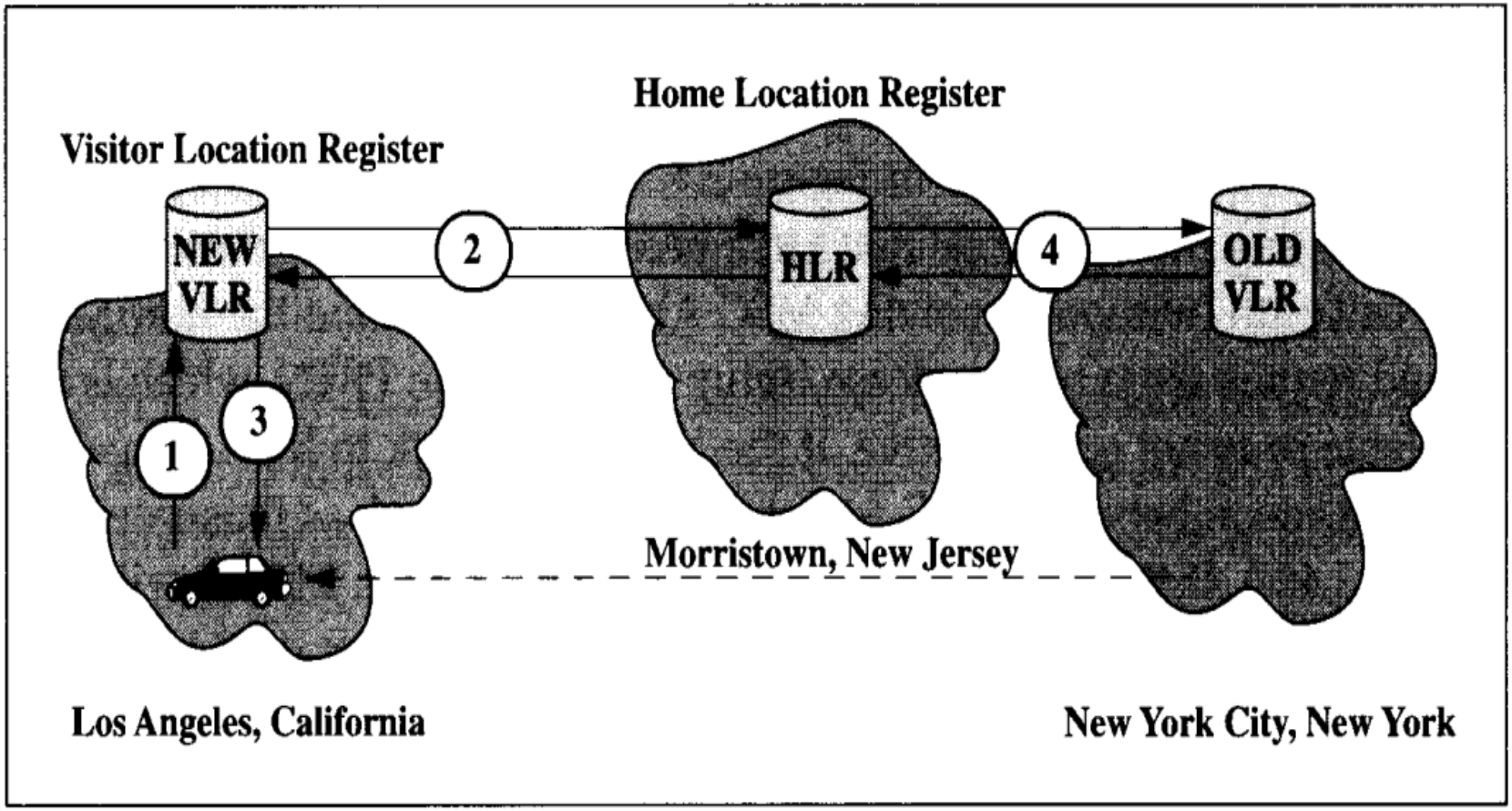


Figure 2.5 MS registration process.

Registration Procedure

- Step 1. Suppose that the home system of a mobile user is in Morristown. When the mobile user moves from one visited system (e.g., New York City) to another (e.g., Los Angeles), it must register in the VLR of the new visited system.
- Step 2. The new VLR informs the mobile user's HLR of the person's current location-the address of the new VLR. The HLR sends an acknowledgment, which includes the MS's profile, to the new VLR.
- Step 3. The new VLR informs the MS of the successful registration.
- Step 4. After step 2, the HLR also sends a **deregistration message** to cancel the obsolete location record of the MS in the old VLR. The old VLR acknowledges the deregistration.

Call delivery procedure

- To originate a call, the MS first contacts the MSC in the visited PCS network.
- The call request is forwarded to the VLR for approval.
- If the call is accepted, the MSC sets up the call to the called party following the standard PSTN call setup procedure.

Call delivery procedure

- **Step 1** . If a wireline phone attempts to call a mobile subscriber, the call is forwarded to a switch, called the **originating switch** in the PSTN, which queries the HLR to find the current VLR of the MS. (1) The HLR queries the VLR in which the MS resides to get a routable address. (2) If the originating switch is not capable of querying the HLR (i.e., it is not equipped to support mobility), the call is routed through the PSTN to the subscriber's gateway MSC, which queries the HLR to determine the current VLR serving the MS.
- **Step 2.** The VLR returns the routable address to the originating switch through the HLR.
- **Step 3**. Based on the routable address, a trunk (voice circuit) is set up from the originating switch to the MS through the visited MSC.

Call delivery procedure

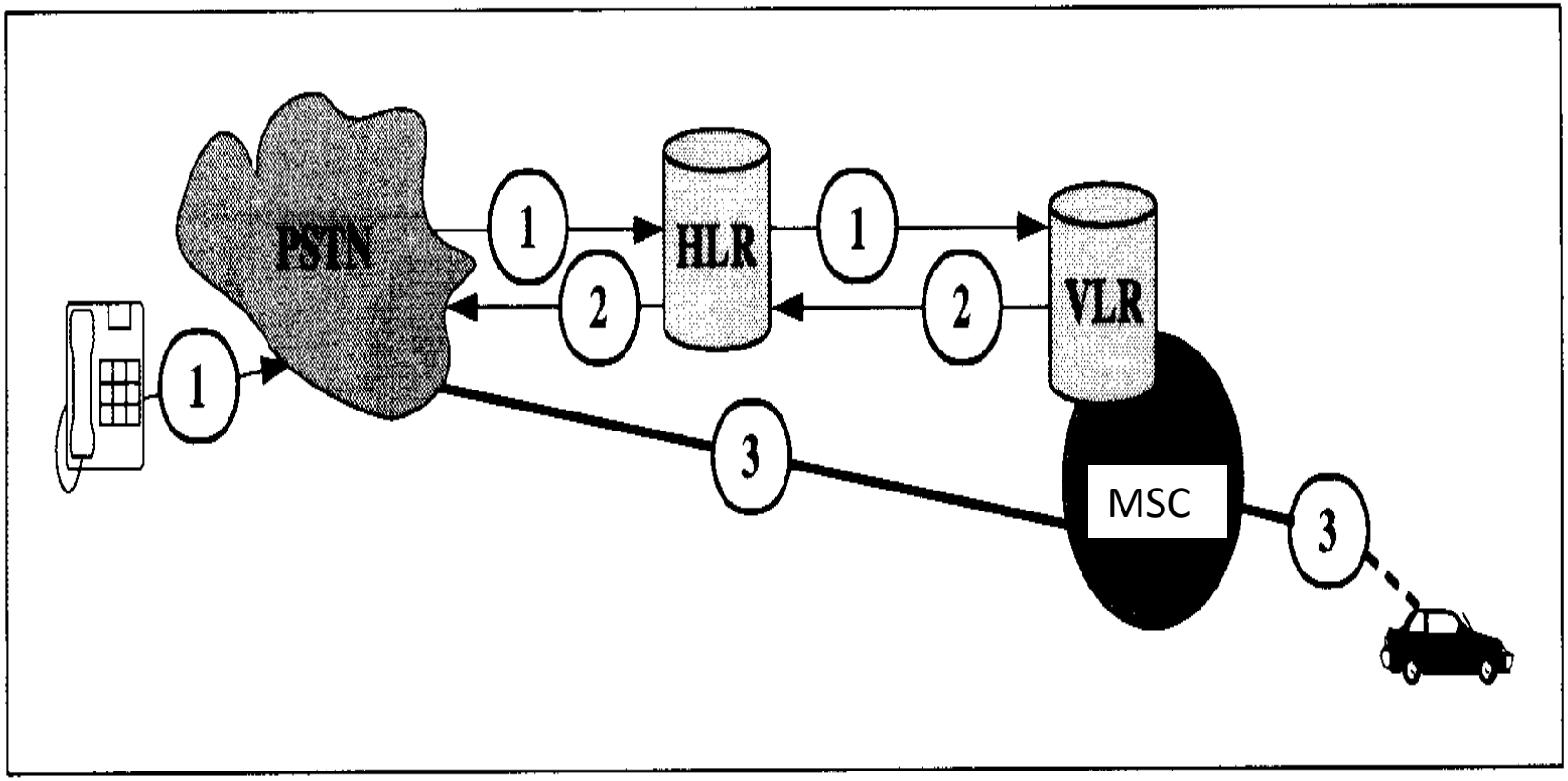


Figure 2.6 Call delivery procedure.

Network Signalling

- The missing parts in the picture are the interactions between the PCS network and the PSTN.
- This section briefly describes how mobile roaming is managed by the PSTN signaling.
- Roaming Management under SS7

Common channel signaling (CCS)

- Common channel signaling (CCS) is a signaling method that provides control and management functions in the telephone network.
- CCS consists of
  - supervisory functions
  - addressing
  - call information provisioning

CCS

- A CCS channel conveys messages to
  - initiate and terminate calls
  - determines the status of some part of the network
  - controls the amount of traffic allowed.
- CCS uses a separate **out-of-band signaling** network to carry signaling messages.

SS7

- Signalling System No. 7 (SS7) is a CCS system developed to satisfy the telephone operating companies' requirements for an improvement to the earlier signaling systems, which lacked the sophistication required to deliver much more than plain old telephone service (POTS).
- Signaling between a PCS network and the PSTN are typically achieved by the SS7 network.

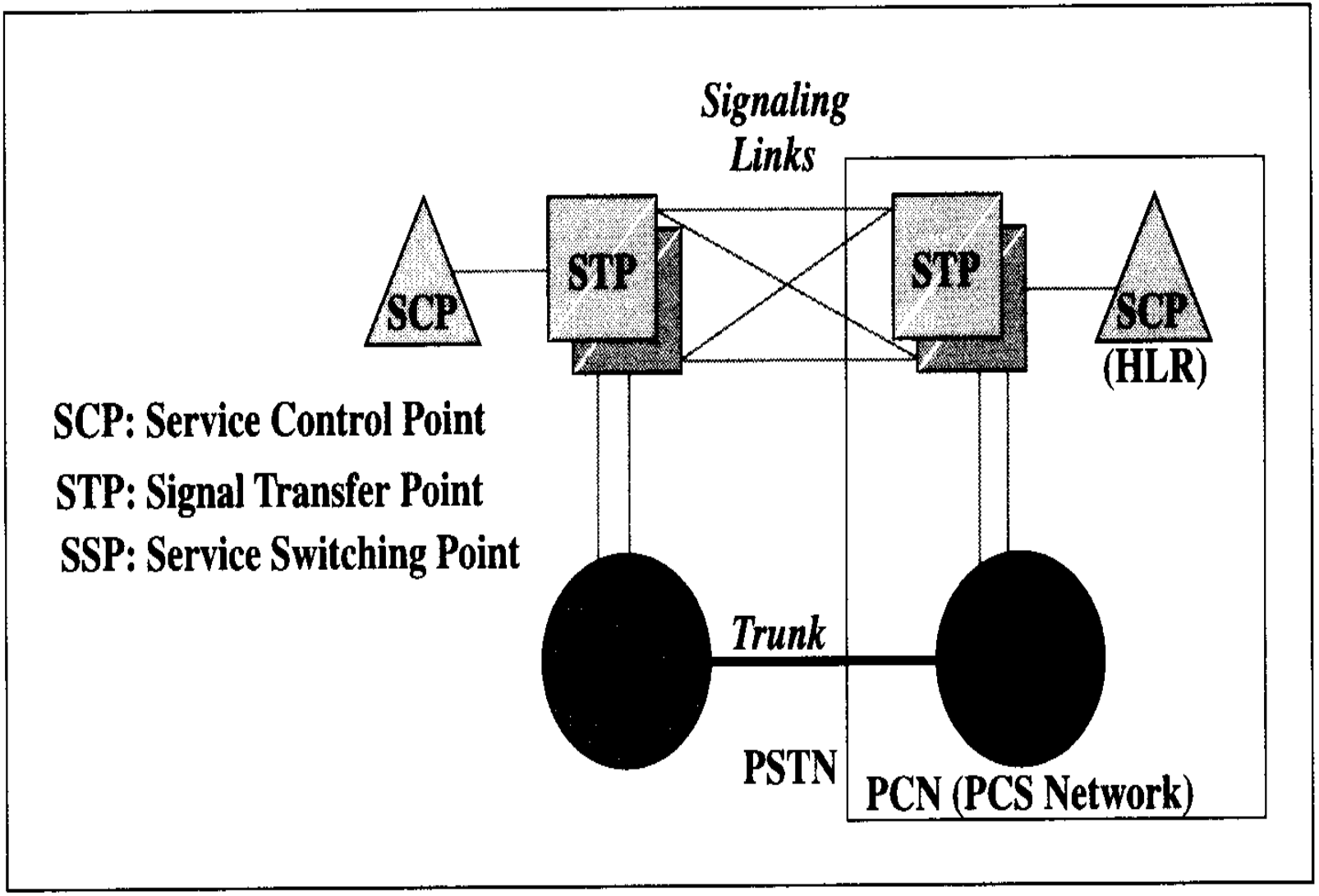


Figure 2.7 Interconnection between a PCS network and the PSTN.

SS7

- Figure shows the network elements that are involved in the interconnection between a PCS network and the PSTN. In the figure, the dashed lines represent the signaling links; the solid line represents a trunk.

SS7

- The SS7 network consists of three distinct components:
  - **Service Switching Point (SSP)**
  - **Signal Transfer Point (STP)**
  - **Service Control Point (SCP)**

Service Switching Point (SSP)

- A telephone switch interconnected by SS7 links. The SSPs perform call processing on calls that originate, tandem, or terminate at that node.
- A local SSP in the PSTN can be a central office (CO) or end office (EO).
- An SSP in a PCS network is called a mobile switching center (**MSC**).

Signal Transfer Point (STP)

- A switch that relays SS7 messages between network switches and databases. Based on the address fields of the SS7 messages, the STPs route the messages to the correct out-going signaling links. To meet the stringent reliability requirements, STPs are provisioned in mated pairs, as shown in Figure.

Service Control Point (SCP)

- Contains databases for providing enhanced services. An SCP accepts queries from an SSP and returns the requested information to the SSP .
- In mobile applications, an SCP may contain an HLR or a VLR.

SS7

- In this network, the trunks (voice circuits) connect SSPs to carry user data/voice information.
- The **signaling links** connect SCPs to STPs, and STPs to SSPs.
- The SSPs and SCPs are connected indirectly through STPs.

Registration

- In this example, the MS moves from VLR1 to VLR2.
- Step 1.
  - The MS enters the area controlled by MSC2.
  - MSC2 launches a registration query to its VLR through STP2, assuming that VLR2 and MSC2 are not co-located.

Registration

- Step 2.
  - VLR2 sends a registration message to the MS's HLR (HLR4 in Figure 2.8).
  - VLR2 may not know the actual address of HLR. Instead, VLR2 sends the message containing the MS identity, called the **Mobile Identification Number (MIN)**, to an **STP (STP3** in our example) that can translate the MIN into the HLR address.
- Step 3.
  - The MIN-to-HLR address translation is performed at STP3 by a table-lookup technique called **global title translation (GTT)**. STP3 then forwards the registration message to HLR.

Registration

- Step 4.
  - After the registration, HLR sends an acknowledgment back to VLR2.
  - Since the address of VLR2 is known, the acknowledgment may be sent to VLR2 using a shortcut, without passing through STP3.
- Step 5.
  - After step 3, HLR sends a deregistration message to VLR1 to cancel the obsolete record.
  - VLR1 then acknowledges the cancellation (not shown in Figure 2.8).

Registration

- In steps 2, 3, 4, and 5, the messages may visit several STPs before arriving at their destinations, and the registration process may generate considerable traffic in the SS7 network.
- Thus, it is desirable to reduce the registration traffic.
- Two approaches have been proposed to reduce the "cost" of deregistration at step 5 in Figure 2.8:
  - **implicit deregistration**
  - **periodic re-registration**

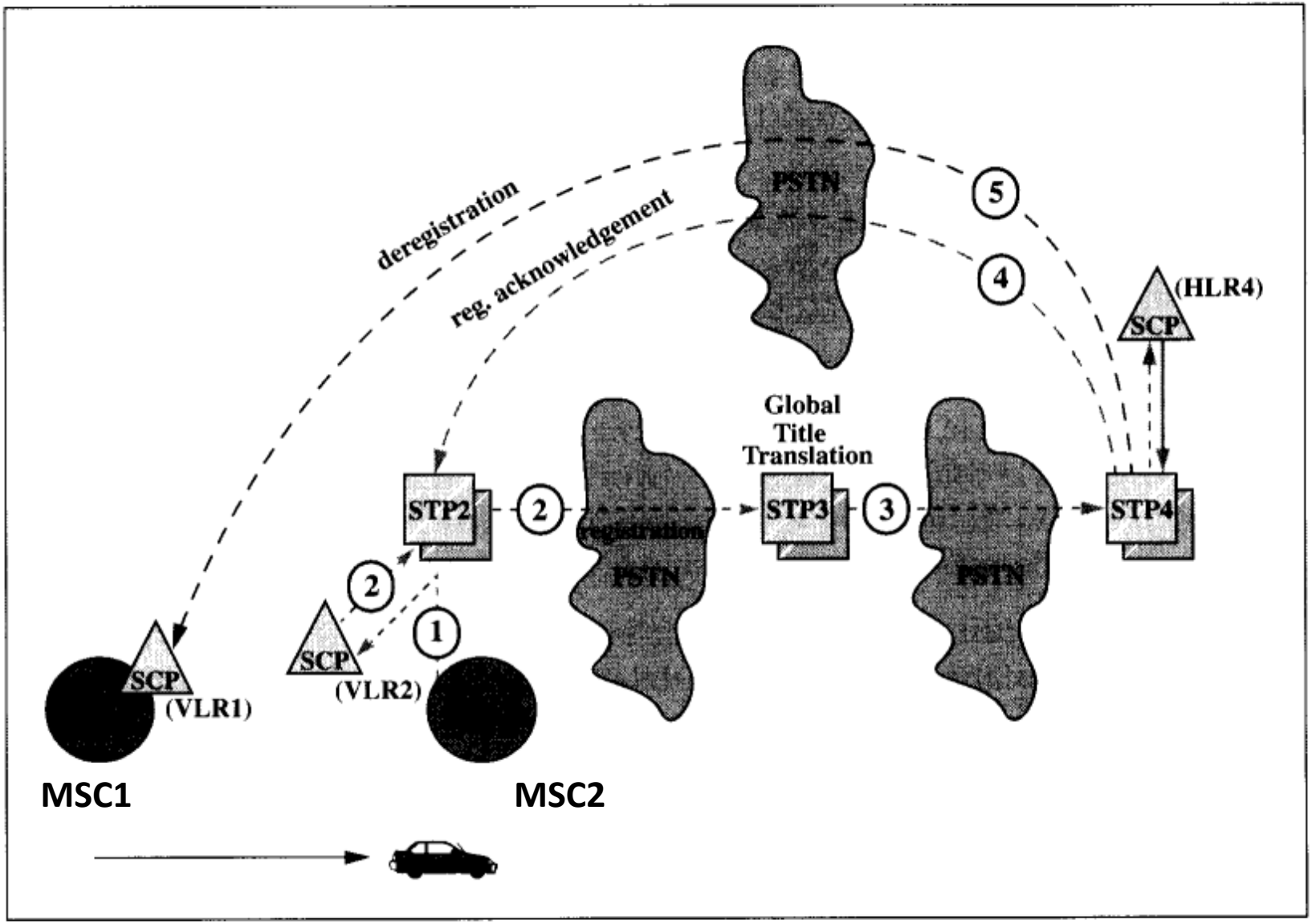


Figure 2.8 Registration through SS7.

Implicit deregistration

- Obsolete VLR records are not deleted until the database is full.
- If the database is full when an MS arrives, a record is deleted, freeing storage space to accommodate the newly arrived MS.
- A replacement policy is required to select a record for replacement (it is possible that a valid record is replaced, and the information is lost).
- Advantage: no deregistration messages are sent among the SS7 network elements.

Periodic re-registration

- The MS periodically reregisters to the VLR.
- If the VLR does not receive the re-registration message within a timeout period, the record is deleted.
- This approach only creates local message traffic between the MSC and the VLR . Furthermore, no SS7 signaling messages are generated if the VLR is co-located with the MSC.

Pointer Forwarding Scheme

- To reduce the registration traffic at steps 2 and 3 in Figure 2.8, a **pointer forwarding scheme** was proposed, which consists of two operations:
- Move operation (registration).
- Find operation (call delivery).

Move operation (registration)

- When an MS moves from one VLR to another, **a pointer** is created from the old VLR to the new VLR. No registration to the HLR is required (see Figure 2.9(a)).

Find operation (call delivery)

- When the HLR attempts to locate the MS for call delivery, the pointer chain is traced. After the find operation, the HLR points directly to the destination VLR (see Figure 2.9(b)).

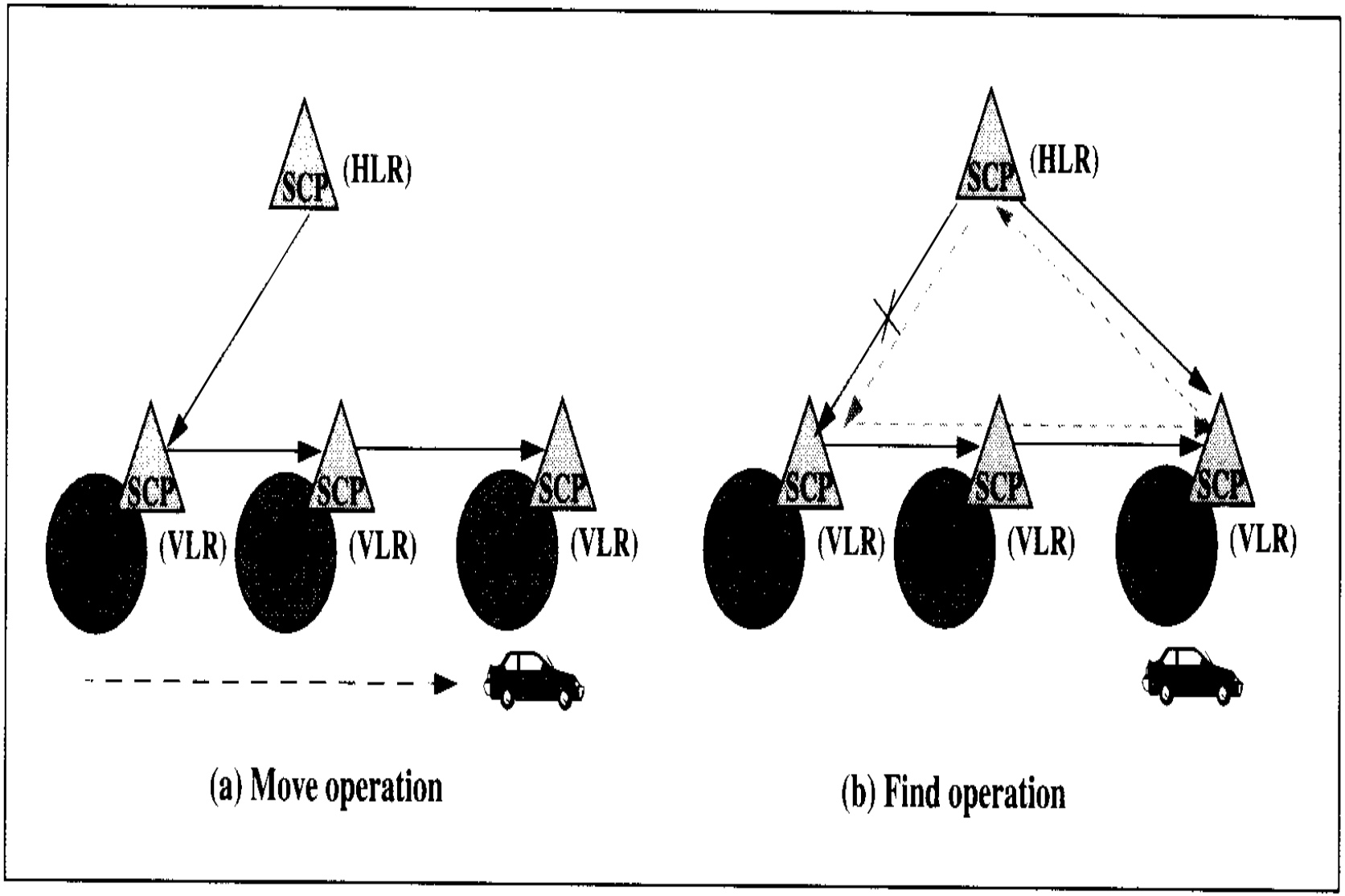


Figure 2.9 Pointer forwarding scheme.

Call Delivery

- Depending on the memory capacities of the VLRs, the pointers in the obsolete chain may or may not be deleted.
- To limit the pointer traversal time in the find operation, the registration procedure in Figure 2.8 may be performed for every k move operations.
- In other words, the number of pointers visited in the find operation will be limited by k. The pointer forwarding scheme should not be considered when the new cost of pointer creation and pointer traversal is higher than the cost of accessing the HLR.
- As performance studies indicate, the pointer forwarding scheme significantly reduces the network traffic in many cases.

Call Delivery

- Similar to the registration process, visits to several STPs and a GTT may be required to access the HLR in call delivery.
- Several STPs may be visited to obtain the routable address from the VLR.
- To reduce the call delivery traffic, a cache scheme was proposed to maintain a cache in the originating SSPs.
- Another possibility is to maintain the cache in the STP that performs GTTs, that is, STP3 in Figure 2.11.
- A cache entry consists of two fields: the **MIN of an MS** and **the address of the current visited VLR of the MS**. The cache contains entries for MSs recently accessed from the SSP

Cache Scheme

- When the calling party originates a call to an MS, the SSP first checks if the cache entry for the MS exists. There are three possibilities:
- Case 1: The cache entry does not exist. The call delivery procedure illustrated in Figure 2.10 is performed.
- Case 2: The cache entry exists and is current. The VLR is directly accessed as shown in Figure 2.11.
- Case 3: The cache entry exists but is obsolete. The procedure detects that the cache entry is obsolete if the queried VLR's response is negative. The call delivery procedure illustrated in Figure 2.10 is performed.

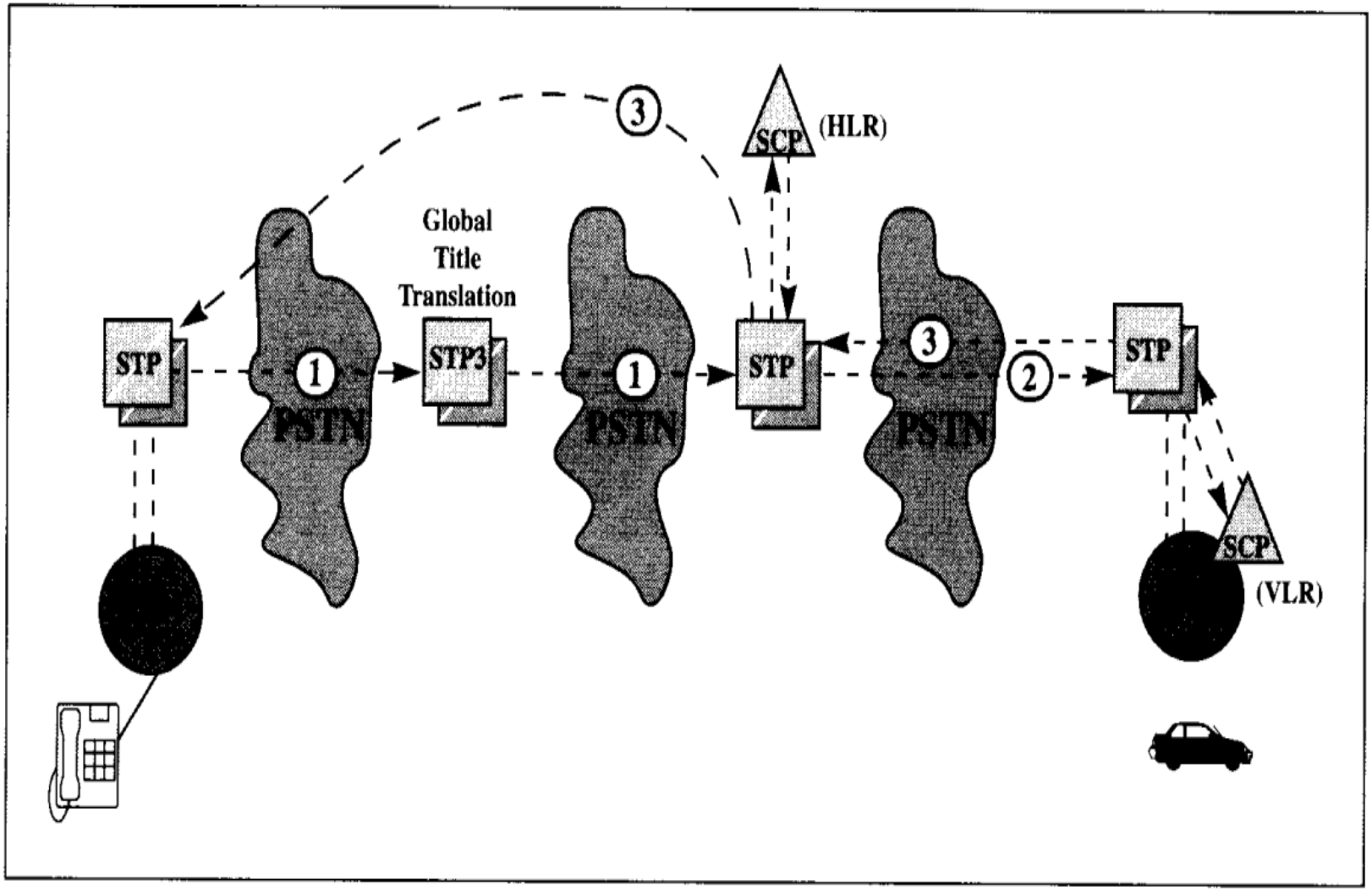


Figure 2.10 Call delivery through SS7.

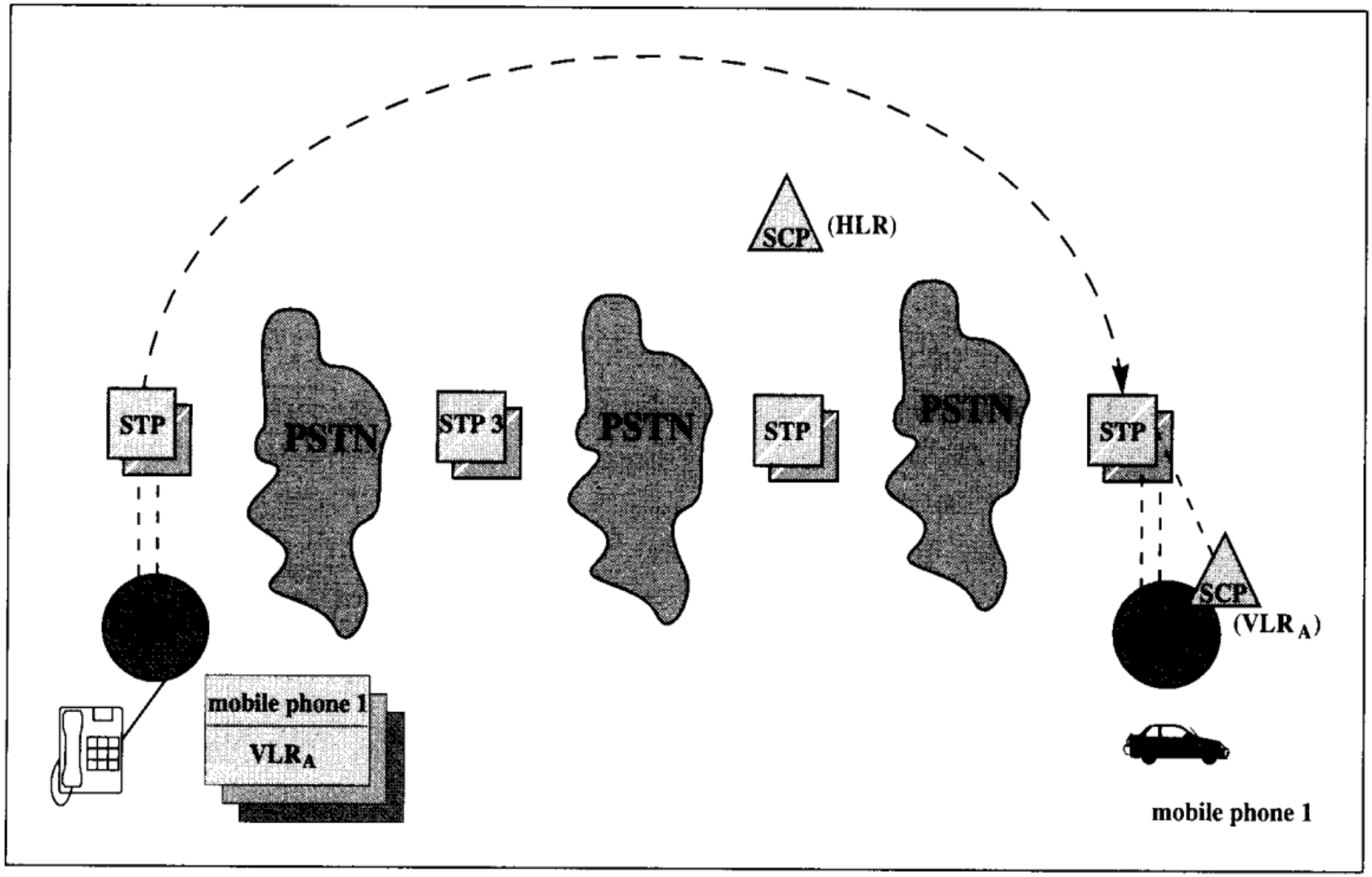


Figure 2.11 Cache scheme.

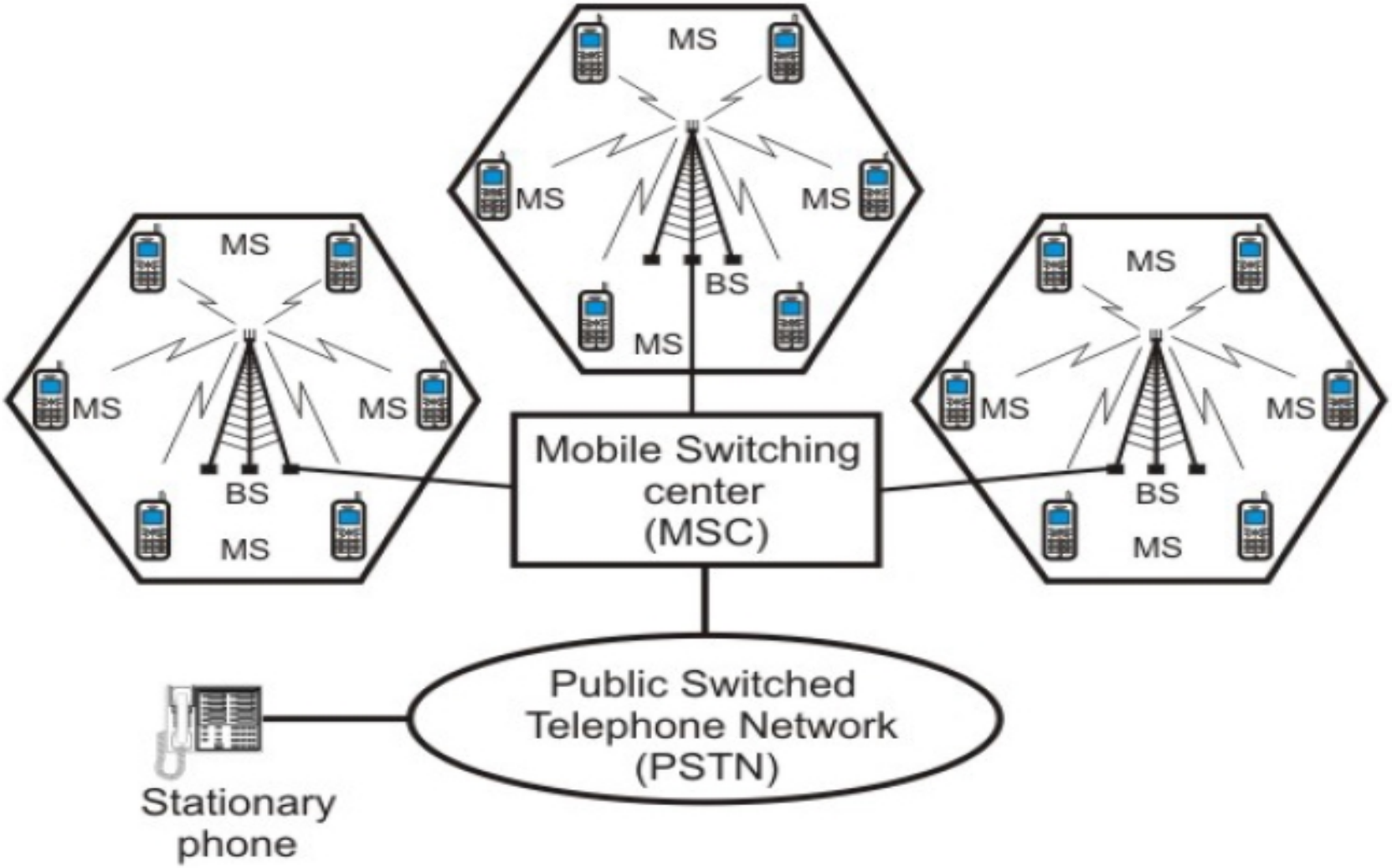
Cellular systems –Basic concepts

- In the early years of mobile radio systems, a large coverage was achieved by using a single high-powered transmitter with the antenna mounted on tall tower. Although a large coverage could be attained by this approach, it does not allow the reuse of the same radio frequencies due to interference.
- The *cellular concept was invented in solving the spectral congestion and user capacity. Cellular telephony is a system-level concept, which replaces a single high power transmitter with a large number of low-power transmitters for communication between any two devices over a large geographic area.*
- *Primary goal of the cellular telephone network is to provide wireless communication between two moving devices, called mobile stations or between one mobile unit and a stationary unit, commonly referred to as land-line unit.*
- *To accommodate a large number of users over a large geographic area, the cellular telephone system divides a large geographical area into small geographical regions called cells. Each cell has a low-power wireless transmitters.*
- *As mobile users travel from cell to cell, their conversations are handed off between cells. Channels (frequencies) used in one cell can be reused in another cell some distance away, which allows communication by a large number stations using a limited number of radio frequencies. To summarize, the basic concept of reuse allows a fixed number of channels to serve an arbitrarily large number of users.*

Cellular systems –Basic components

A cellular system comprises the following basic components:

- Mobile Stations (MS): Mobile handsets, which is used by an user to communicate with another user
- Cell: Each cellular service area is divided into small regions called cell (5 to 20 Km)
- Base Stations (BS): Each cell contains an antenna, which is controlled by a small office.
- Mobile Switching Center (MSC): Each base station is controlled by a switching office, called mobile switching center



Schematic diagram of a cellular telephone system

Forward and Reverse Channels

- Forward voice channel (FVC): is used for voice transmission from the base station to the mobile station.
- Reverse voice channel (RVC): is used for voice transmission from the mobile station to the base station.
- However, as we know voice channel can be only activated once the hand shaking has been done. So we required some kind of control channel.
- A forward control channel (FCC): is used for initiating a call from the base station to the mobile station.
- Reverse control channel (RCC): is used to initiate a call from the mobile station to the base station.
- Please remember the FCC and the RCC, the control channels are also called setup channels because of the nature of work they do.

Anatomy of a cellular call.

- A cell phone when turned on but not engaged in a call scans the group of forward control channels to determine the one with the strongest signal. (Please note even though you are not making a call, you are using a battery power. Therefore when you buy a phone, you have two kinds of time available; the talk time and the standby time. So your phone can get discharged over several days even though you don't make a single call because it has to monitor and keep itself updated as to which cell it is talking to.)
- Therefore even though you don't touch the phone, the display may change depending on which base station it is connected to.
- The mobile phone monitors the channel and keeps on monitoring because if the strength drops below a certain threshold, it scans for the next strongest.
- Control channels are defined and standardized over the entire area of service. Typically the control channels use up to 5 % of the total number of channels.

Cellular system- Operation

- Basic operations of transmitting and receiving in a cellular telephone network are discussed in this section.
- Transmitting involves the following steps:

1. A caller enters a 10-digit code (phone number) and presses the send button.
2. The MS scans the band to select a free channel and sends a strong signal to send the number entered.
3. The BS relays the number to the MSC.
4. The MSC in turn dispatches the request to all the base stations in the cellular system.
5. The Mobile Identification Number (MIN) is then broadcast over all the forward control channels throughout the cellular system. It is known as *paging.*
6. The MS responds by identifying itself over the reverse control channel.
7. The BS relays the acknowledgement sent by the mobile and informs the MSC about the handshake.
8. The MSC assigns an unused voice channel to the call and call is established.

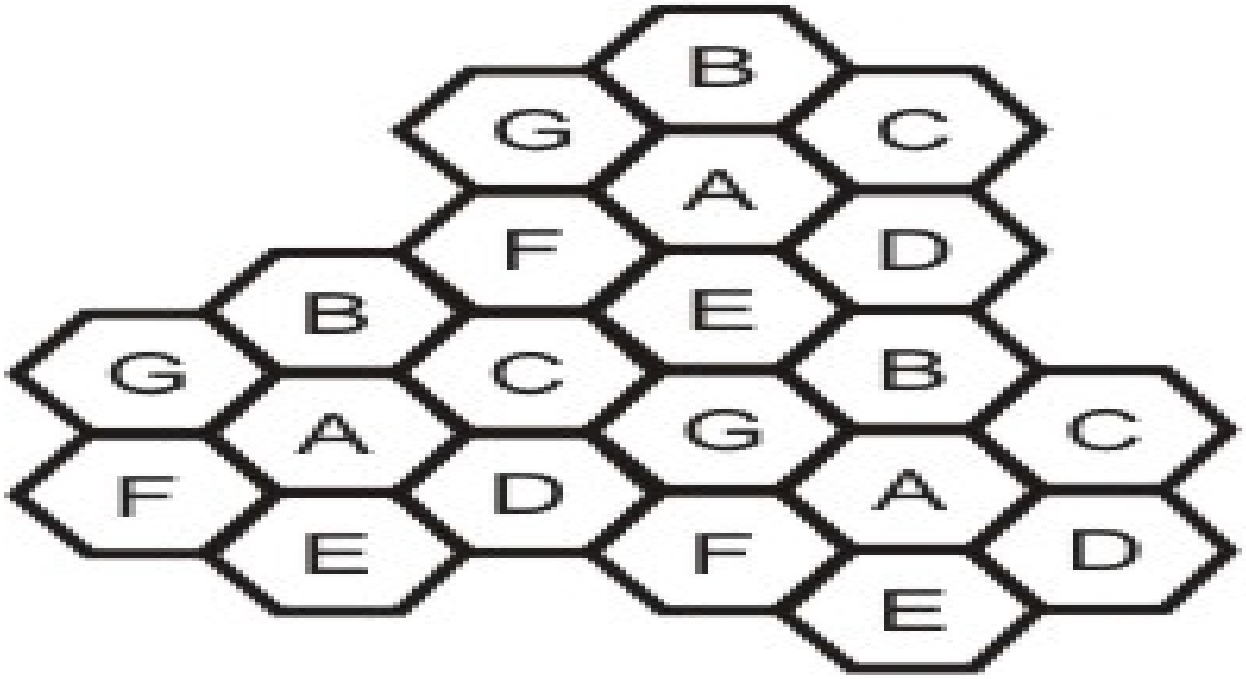
Frequency reuse- Need

- Suppose we had fixed telephone networks and they were running wires to every household but we are not talking about wire line truly. We have compared it to a wireless situation because we are looking at wireless cellular systems. Suppose we would like to give every household in Delhi a voice bandwidth. They can talk. So we need to give about 4 KHz of spectrum but the number of households in Delhi is about 12.5 million. Suppose we get into a stage where we can actually give them 4KHz of bandwidth, we are talking about (**12.5 million* 4 KHZ = 50GHz**) 50GHz of bandwidth. Clearly we cannot allow this kind of outrage on a mobile phone network. We do not have the luxury to touch fifty GHz of bandwidth. We have to reuse our frequency. So clearly in practical, no other service is possible using radio transmission. Most of the spectrum unfortunately will remain unused most of the time. If I give a phone line connection, I cannot do adaptive reallocation. So clearly there is a need to do frequency reuse.

Frequency Reuse Concept

- Cellular radio systems rely on intelligent allocation and reuse of channels throughout the coverage area.
- Each base station is allocated a group of radio channels to be used within the small geographic area which we have already defined as a cell. So it is not just a frequency but a frequency band.
- Neighboring base stations are given different channel allocation from each other to avoid interference.
- By the design of antennas and regulating the transmit power, the coverage area within the cell is limited. The same group of frequencies are reused to cover another cell separated by a large enough distance to keep this co-channel interference which is generated by cells using the same frequency band under control.
- The design procedure for allocating channel for the cellular base station within a system is called the frequency reuse or frequency planning

Frequency Reuse Concept



Multiple access Techniques

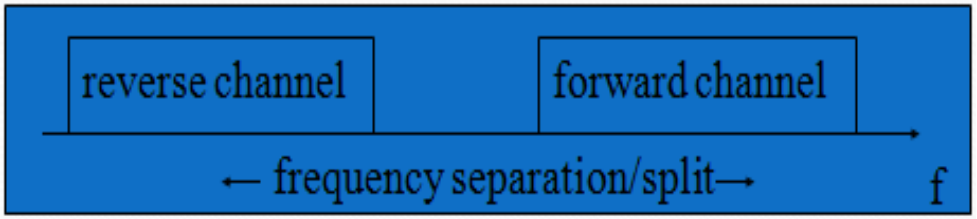
- Multiple access techniques are used to **allow a large number of mobile users to share the allocated spectrum** in the most efficient manner.
- As the spectrum is limited, so the sharing is **required to increase the capacity** of cell or over a geographical area by allowing the available bandwidth to be used at the same time by different users.
- And this must be done in a way such that the quality of service **(QOS) doesn't degrade** within the existing users.

Multiple access Techniques

- In wireless communication systems it is often desirable to allow the subscriber to send simultaneously information to the base station while receiving information from the base station
- Duplexing is needed to allow subscribers send and receive information simultaneously. e.g., telephone systems,
- Duplexing may be done using
  - frequency domain technique
  - time domain technique

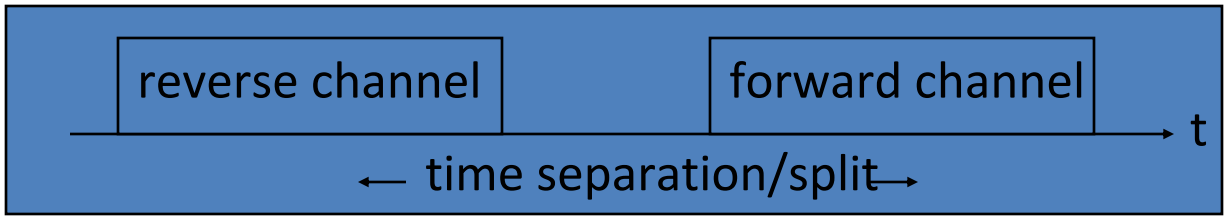
Frequency division duplexing (FDD)

- FDD provides two distinct bands of frequencies for every user
  - forward band (provides traffic from Base station to mobile unit)
  - reverse band (for traffic from mobile unit to Base station)
- Any duplex channel Consists of two simplex channels (a forward and reverse) and a device called duplexer is used inside each mobile unit and base station
- frequency separation between forward band and reverse band is constant throughout the system



Time division duplexing (TDD)

- TDD uses different time slots instead of frequency to provide a forward and reverse link.
- In TDD, multiple users share a single radio channel by taking turns in time domain
  - forward time slot
  - reverse time slot
- no duplexer is required (a simple switch can be used)
- Communication is not full-duplex



Techniques for Multiple Access

- Three major techniques:
  - Frequency division multiple access (FDMA)
  - Time division multiple access (TDMA)
  - Code division multiple access (CDMA)
- Others:
  - Packet radio (PR)
  - Space division multiple access (SDMA)

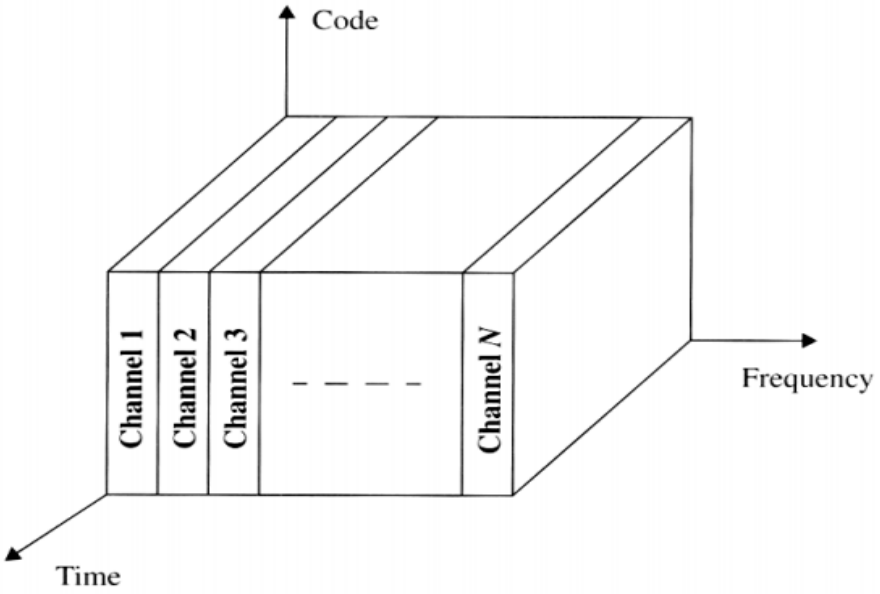
- These techniques can be grouped as **narrowband** and **wideband** systems, depending upon how the available bandwidth is allocated to the users.
- The duplexing technique of a multiple access system is usually described along with the particular multiple access scheme

Narrowband and **Wideband** systems

- Narrowband systems
- Bandwidth of the signal is narrow compared with the coherence bandwidth of the channel
- In NB systems available radio spectrum is divided into large number of narrowband channels usually FDD (large frequency split)
- Wideband system
- The transmission BW of a single channel is much larger than the coherence bandwidth of the channel
- users are allowed to transmit in a large part of the spectrum
- large number of transmitters on one channel

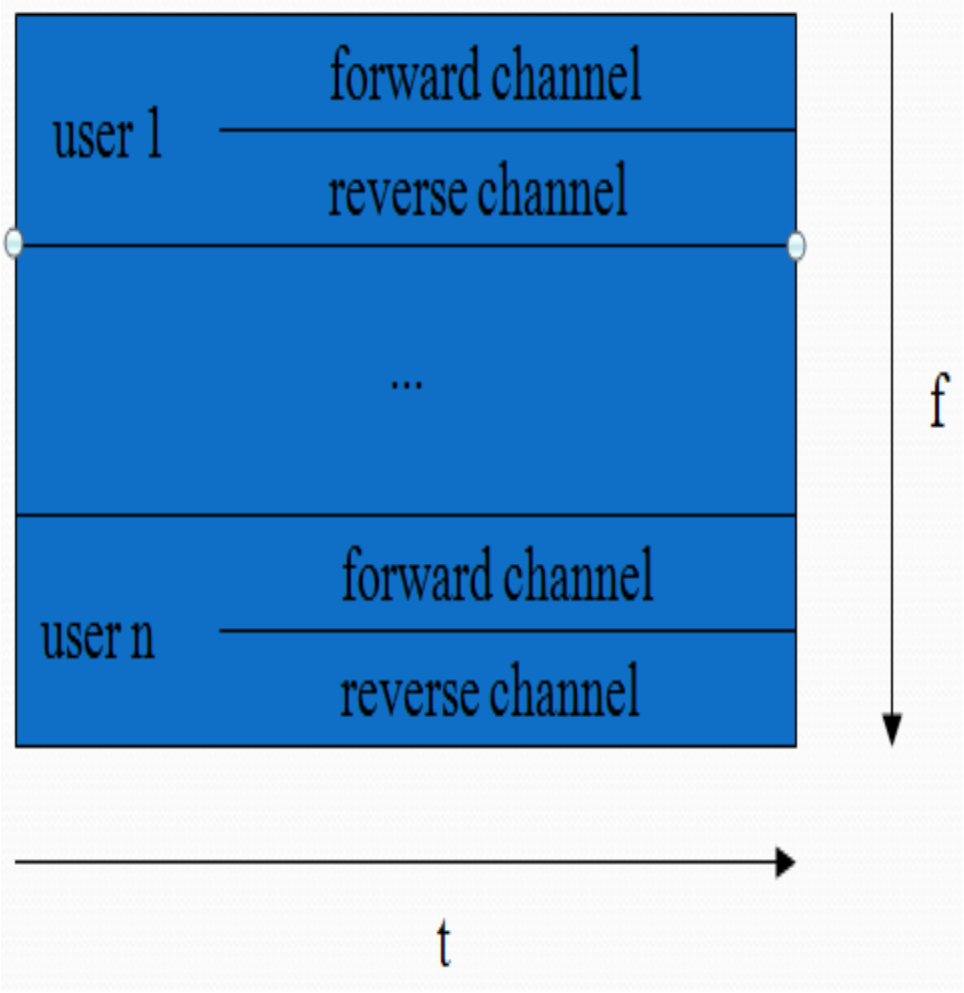
Frequency division multiple access (FDMA)

- Each user is allocated a unique frequency band or channel.
- These channels are assigned on demand, and can not be shared

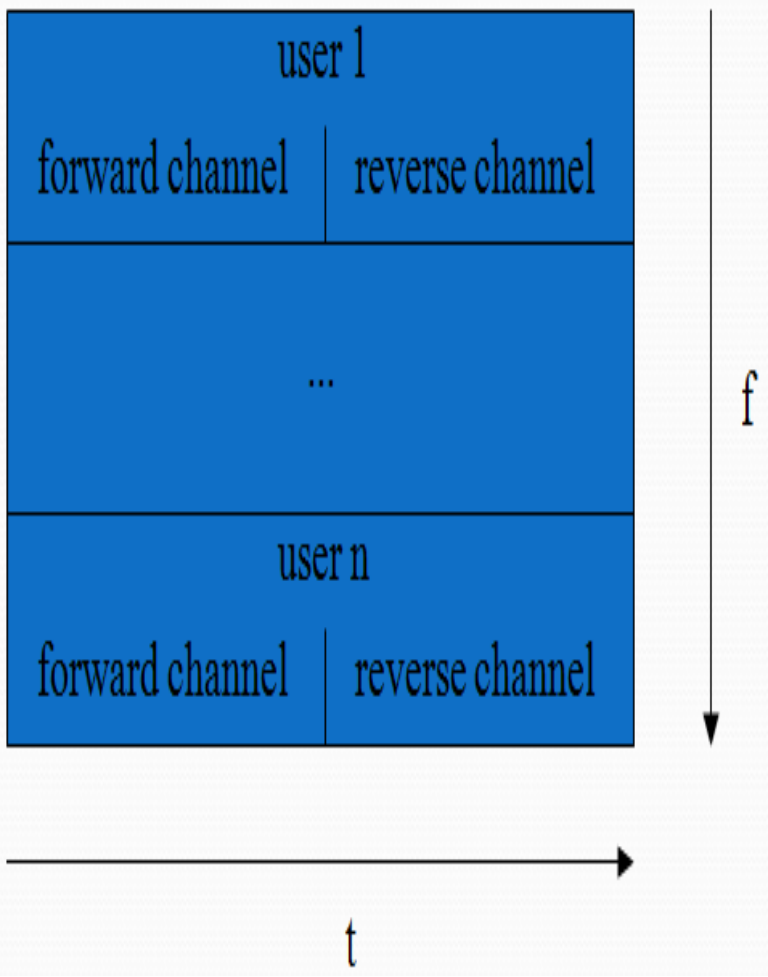


FDMA where different channels are assigned different frequency bands.

FDMA/FDD



FDMA/TDD



FDMA - characteristics

- The FDMA channel carries only **one phone circuit at a time**.
- If an FDMA channel is not in use, then **it sits idle** and cannot be used by other users to increase or share capacity. It is essentially a **wasted resource**.
- After the assignment of a voice channel, the base station and the mobile transmit simultaneously and continuously.
- FDMA requires **tight RF filtering** to minimize adjacent channel interference.

FDMA - Features

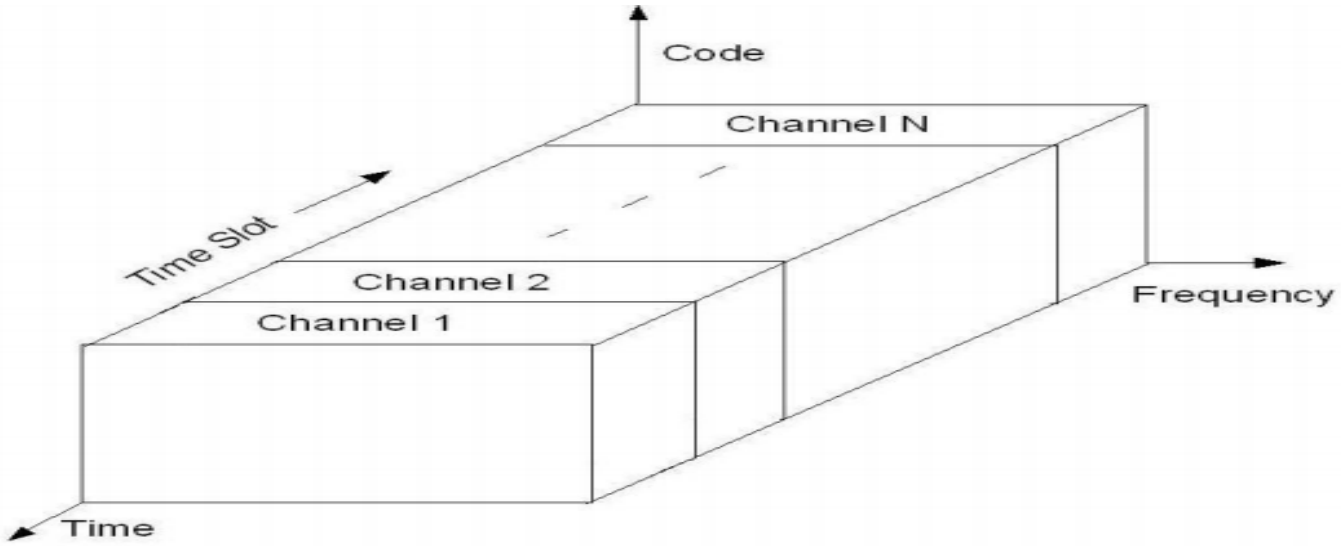
- The bandwidths of FDMA channels are relatively narrow (30 kHz) as each channel supports only one circuit per carrier. That is, FDMA is usually **implemented in narrowband systems**.
- Since FDMA is a continuous transmission scheme, **fewer bits are needed for overhead** purposes (such as synchronization and framing bits) as compared to TDMA.
- The FDMA mobile unit **uses duplexers** since both the transmitter and receiver operate at the same time. This results in an increase in the cost of FDMA subscriber units and base stations.

Time division multiple access (TDMA

- In digital systems, continuous transmission is not required because users do not use the allotted bandwidth all the time.
- In such cases, TDMA is a **complimentary access technique to FDMA**. Global Systems for Mobile communications (GSM) uses the TDMA technique.

TDMA

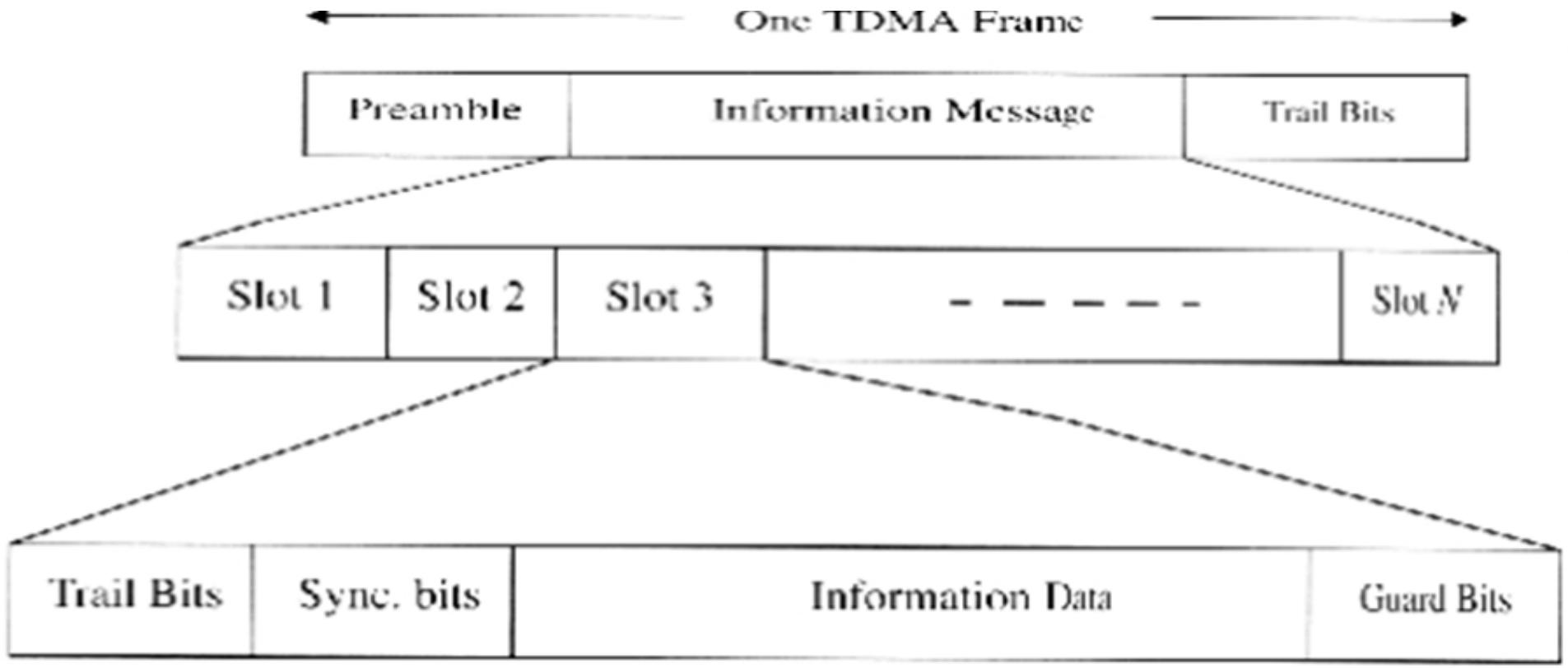
- In TDMA, the entire **bandwidth is available** to the user **but only for a finite period** of time.
- Transmit data in a **buffer-and-burst method,** the transmission for any user is non continuous



TMDA Frame structure

- Frame ---- consists of a **number of slots** (information message), together **with a preamble, and tail bits**.
- Preamble ---- contains **the address and synchronization information** that both the base station and the subscribers use to identify each other.
- Guard times ---- allow synchronization of the receivers between different slots and frames.

TDMA Frame structure



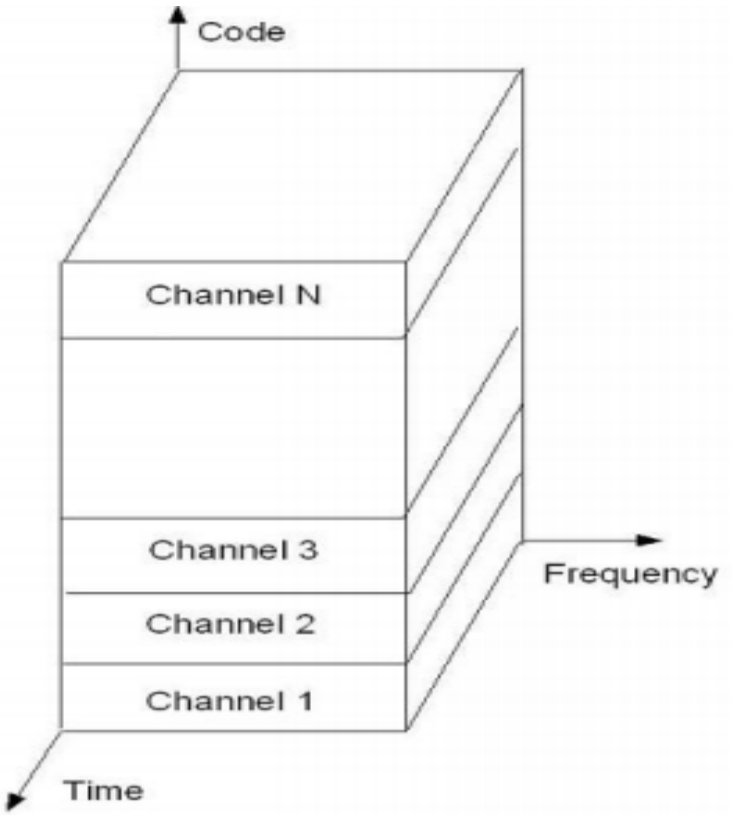
TDMA frame structure. The frame is cyclically repeated over time.

TDMA - Features

- TDMA **shares a single carrier frequency with several users** where each users makes use of non overlapping time slots. The number of time slots per frame depends on several factors such as modulation technique, available bandwidth etc.
- Data transmission in TDMA is not continuous but **occurs in bursts**. This results in **low battery consumption** since the subscriber transmitter can be turned OFF when not in use.
- Because of a discontinuous transmission **in TDMA the handoff process is much simpler for a subscriber unit,** since it is able to listen to other base stations during idle time slots.
- TDMA uses different time slots for transmission and reception thus **duplexers are not required**.
- TDMA has an advantage that is possible to allocate different numbers of time slots per frame to different users. Thus **bandwidth can be supplied on demand to different users by concatenating or reassigning time slot based on priority**

Code division multiple access (CDMA)

- In CDMA, the same bandwidth is occupied by all the users, however they are all assigned separate codes, which differentiates them from each other.
- CDMA utilize a spread spectrum technique in which a spreading signal (which is uncorrelated to the signal and has a large bandwidth) is used to spread the narrow band message signal.



CDMA - features

- Many users of a CDMA system share the same frequency for complete time. Either TDD or FDD may be used.
- Unlike TDMA or FDMA, CDMA has a **soft capacity limit**. Increasing the number of users in a CDMA system raises the noise floor in a linear manner. Thus, there is no absolute limit on the number of users in CDMA. Rather, the system performance gradually degrades for all users as the number of users is increased, and improves as the number of users is decreased.
- Multipath fading may be substantially reduced because the signal is spread over a large spectrum.

Approach	SDMA	TDMA	FDMA	CDMA
Idea	Segment space into cells/sectors	Segment sending time into disjoint time-slots, demand driven or fixed patterns	Segment the frequency band into disjoint sub-bands	Spread the spectrum using orthogonal codes
Terminals	Only one terminal can be active in one cell/one sector	All terminals are active for short periods of time on the same frequency	Every terminal has its own frequency, uninterrupted	All terminals can be active at the same place at the same moment, uninterrupted
Signal separation	Cell structure directed antennas	Synchronization in the time domain	Filtering in the frequency domain	Code plus special receivers

Approach	SDMA	TDMA	FDMA	CDMA
Advantages	Very simple, increases capacity per km^2	Established, fully digital, very flexible	Simple, established, robust	Flexible, less planning needed, soft handover
Disadvantages	Inflexible, antennas typically fixed	Guard space needed (multi-path propagation), synchronization difficult	Inflexible, frequencies are a scarce resource	Complex receivers, needs more complicated power control for senders
Comment	Only in combination with TDMA, FDMA or CDMA useful	Standard in fixed networks, together with FDMA/SDMA used in many mobile networks	Typically combined with TDMA (frequency hopping patterns) and SDMA (frequency reuse)	Used in many 3G systems, higher complexity, lowered expectations; integrated with TDMA/FDMA